

This comprehensive overview covers the fundamental concepts of the Cisco IOS Command-Line Interface (CLI) as presented in the source material.

Introduction to Cisco IOS and the CLI

Cisco IOS (Internetwork Operating System) is the operating system used on Cisco devices like routers, switches, and firewalls. The **CLI** (Command-Line Interface) is the text-based interface used to configure these devices, which is generally preferred by network engineers over a **GUI** (Graphical User Interface).

Connecting to a Cisco Device

To initially configure a device, you must connect to its **console port**.

- **Physical Ports:** Devices typically have an RJ45 console port or a USB mini-B connector.
- **Cables:** An RJ45 console connection requires a **rollover cable**, which has eight pins wired in reverse (pin 1 to pin 8, pin 2 to pin 7, etc.).
- **Adapters:** Since most modern laptops lack serial ports, a DB9-to-USB adapter is often necessary.
- **Terminal Emulators:** Software like **PuTTY** is used to access the CLI. The default connection settings for Cisco devices are:
 - **Speed (Baud rate):** 9600 bps
 - **Data bits:** 8
 - **Stop bits:** 1
 - **Parity:** None
 - **Flow control:** None

CLI Operating Modes

The CLI uses different modes, each providing different levels of access:

1. **User EXEC Mode:** Indicated by the `>` prompt (e.g., `Router>`). This mode is limited and used primarily for basic monitoring.
2. **Privileged EXEC Mode:** Indicated by the `#` prompt (e.g., `Router#`). It provides full access to view configurations and restart the device. You enter this mode by typing the `enable` command.
3. **Global Configuration Mode:** Indicated by the `(config)#` prompt (e.g., `Router(config)#`). This is the mode used to actually change the device's configuration. You enter this by typing `configure terminal` (or the shortcut `conf t`).

CLI Navigation and Shortcuts

Cisco IOS includes several features to make command entry more efficient:

- **Help:** Type a `?` to see available commands or how to complete a specific command.
- **Tab Completion:** Pressing the **Tab** key will automatically complete a partially typed word.
- **Shortcuts:** Commands can be shortened as long as they are unique. For example, `en` works for `enable` and `conf t` works for `configure terminal`.
- **The `do` Command:** You can run Privileged EXEC mode commands (like `show` commands) while in configuration mode by prefixing them with `do` (e.g., `do show run`).

Configuration Management

Cisco devices maintain two separate configuration files:

- **Running-config:** The active configuration currently in the device's RAM. Changes made in the CLI take effect immediately here.
- **Startup-config:** The configuration stored in non-volatile memory that loads when the device boots up.
- **Saving Changes:** To ensure changes persist after a reboot, you must copy the running-config to the startup-config. This can be done using `write`, `write memory`, or `copy running-config startup-config`.

Security and Password Configuration

Protecting access to the CLI is critical for network security.

- `enable password [password]`: Sets a password for Privileged EXEC mode, but displays it in **plaintext** within the configuration file.
- `service password-encryption`: Encrypts all plaintext passwords in the configuration using Cisco's proprietary "Type 7" algorithm. However, this encryption is weak and easily cracked.
- `enable secret [password]`: This is the **most secure method**. It uses MD5 encryption ("Type 5") and is encrypted by default.
- **Precedence:** If both `enable password` and `enable secret` are configured, the device will ignore the `enable password` and only accept the `enable secret`.

Removing Commands

To delete or undo a configuration, you generally type `no` before the original command (e.g., `no service password-encryption`). Note that disabling encryption will not decrypt existing passwords; it only ensures that future passwords remain in clear text.

Cisco IOS CLI Operating Modes

The CLI is organized into different levels of access, each with a specific purpose and distinct prompt.

- **User EXEC Mode (`Router>`):** This is the default mode when first entering the CLI. It is very limited, allowing for basic monitoring but no configuration changes.
 - **Privileged EXEC Mode (`Router#`):** Entered using the `enable` command, this mode provides full access to view configurations and restart the device.
 - **Global Configuration Mode (`Router(config)#`):** Entered using the `configure terminal` command, this is the primary mode used to modify the device's settings.
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Cisco CLI Commands and Functions

The following commands are essential for navigating and managing a Cisco device:

Command	Function
<code>enable</code>	Moves the user from User EXEC mode to Privileged EXEC mode.
<code>configure terminal</code> (shortcut: <code>conf t</code>)	Moves the user from Privileged EXEC mode to Global Configuration mode.
<code>?</code> (Question Mark)	Provides help by displaying available commands or options to complete a specific command.
<code>enable password [password]</code>	Sets a plaintext password to protect access to Privileged EXEC mode.
<code>enable secret [password]</code>	Sets a highly secure, MD5-encrypted password for Privileged EXEC mode. If both are set, the <code>enable secret</code> overrides the <code>enable password</code> .
<code>service password-encryption</code>	Encrypts all plaintext passwords (like the <code>enable password</code>) in the configuration file using a weak Type 7 encryption.
<code>show running-config</code> (shortcut: <code>sh run</code>)	Displays the current active configuration stored in the device's RAM.
<code>show startup-config</code>	Displays the configuration file stored in non-volatile memory that loads when the device boots.

copy running-config startup-config	Saves the active configuration to non-volatile memory so changes persist after a reboot.
write or write memory	Shortcut commands that perform the same function as copy running-config startup-config .
do [command]	Allows the user to run Privileged EXEC commands (such as show) while staying in Global Configuration mode.
no [command]	Used as a prefix to remove or disable a previously entered configuration command.
exit	Moves the user back one level in the mode hierarchy or logs the user out.

CLI Navigation Tips

- **Tab Key:** Automatically completes a word if you have typed enough letters for it to be unique.
- **Shortcuts:** Commands can be abbreviated as long as the abbreviation is unique (e.g., **en** for **enable** or **conf t** for **configure terminal**).
- **Case Sensitivity:** All passwords configured in the CLI are case-sensitive.